An order on the Leech lattice from a $\mathbb{Z}_3$-symmetric triple-octonion product

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Abstract

We show that the Leech lattice Λ admits the structure of an order in a 24-dimensional real algebra: the lattice is closed under a bilinear product on $\mathbb{R}^{24}$, assembled from nine octonion-multiplication blocks under cyclic $\mathbb{Z}_3$ permutation of three octonion factors. In Robert A. Wilson's sublattice description of Λ, closure turns on a transposition σ of two imaginary basis units: σ leaves the E_8 lattice L invariant and is an octonion-algebra isomorphism, but moves Wilson's sublattices $L\bar{s}$ and Ls. The proof reduces to four lemmas, three of which follow from the closure of one maximal order of integral octonions. More generally, for every linear map g with $g(L) = L$, the triple product built from the g-twisted octonion product closes on Λ exactly when Wilson's two sublattices are closed under that product. We determine idempotents, automorphism group, square-zero elements of the algebra, and the index of the span of the products of the order. We discuss the construction in the context of related work. A historical appendix (1923–1946) traces σ to Bruck's repair of Kirmse's integral octonions. The research record of this human–AI collaboration is public.

Keywords: Leech lattice; integral octonions; E_8 lattice; nonassociative algebra; nonunital order.
MSC 2020: Primary 11H06, 17A75; Secondary 17D05, 68V05.

1 Introduction

The Leech lattice is the unique even unimodular lattice of rank 24 with no vectors of squared length 2 [ConwaySloane1999]. Its minimal shell consists of 196,560 vectors of squared length 4, and its automorphism group is the Conway group Co_0, a double cover of the sporadic simple group Co_1. Throughout this paper, Λ denotes the $\sqrt{2}$-scaled copy of this lattice, at the scale of Wilson's construction (Section 2.3); its minimal shell $\mathrm{Min}(\Lambda)$ consists of vectors of squared length 8.

In rank 8, the E_8 lattice L is closed under an octonion multiplication. For a fixed multiplication, seven maximal orders of integral octonions contain all eight basis units, each an E_8 lattice [Coxeter1946, ConwaySmith2003] (Appendix B). Wilson [Wilson2009, Sections 3–4] showed that Λ embeds in L^3 via three sublattice conditions, raising the question: does Λ inherit an algebraic structure from the octonion product?

This question has been studied from several angles. Geoffrey M. Dixon [Dixon2010] obtained the integral octonions from an octonion XY-product and represented Λ in $\mathbb{O}^3$. John C. Baez and Greg Egan [Baez2014] explored Jordan-algebraic structures containing the Leech lattice within the exceptional Jordan algebra $\mathfrak{h}_3(\mathbb{O})$, finding closure under a doubled Jordan product.

Here we introduce a product on the direct sum $\mathbb{O}_1 \oplus \mathbb{O}_2 \oplus \mathbb{O}_3$ of three octonion blocks: it multiplies blocks pairwise with the same octonion multiplication and routes each product to one block by a $\mathbb{Z}_3$-symmetric rule. Applied to vectors of Λ, the product closes once the octonion multiplication is twisted by a transposition σ of two imaginary basis units: we show that, with this twist, Wilson's sublattice conditions yield the required closure inclusions.

Theorem 1.1. *Let $\star$ denote the $\mathbb{Z}_3$-symmetric triple-octonion product on $\mathbb{R}^{24} = \mathbb{O}_1 \oplus \mathbb{O}_2 \oplus \mathbb{O}_3$ of Definition 3.3. Then $u \star v \in \Lambda$ for all $u, v \in \Lambda$ in Wilson's representation [Wilson2009].*

Corollary 1.2. *$(\Lambda, +, \star)$ is an order[1] in the $\mathbb{R}$-algebra $(\mathbb{R}^{24}, +, \star)$.*

Proof. The product $\star$ is bilinear (it is a sum of bilinear octonion products). By Theorem 1.1, $\star$ maps $\Lambda \times \Lambda$ into Λ. □

Remark 1.3 (Order-closure, not shell-closure). The closure $\Lambda \star \Lambda \subseteq \Lambda$ makes $(\Lambda, +, \star)$ an order in the ring-theoretic sense: a subring that is a free $\mathbb{Z}$-module of rank equal to the ambient algebra's dimension and spans it. The product $\star$ does not preserve the minimal shell (Remark 5.1). Closure of the minimal shell, which spans Λ, would be the stronger property.

The scope of this paper is deliberately bounded: we establish existence of the order, with a proof that is elementary (four short lemmas, three of which follow from the closure of one maximal order). Section 2 fixes the octonion multiplication, the E_8 lattice L, the maximal order on which the proof rests, and Wilson's description of Λ. Section 3 defines the product, and Section 4 proves Theorem 1.1. Section 5 determines algebraic properties of the order: that standard identities fail on it; its idempotents; that it is not maximal; its automorphism group; and its square-zero elements. It also characterizes the twists other than σ that close on Λ. Section 6 places the construction among related work, Section 7 concludes with the open questions, and Section 8 is an outlook. Appendix A reads the closure criterion modulo 2, Appendix B traces the integral octonions from Dickson to Coxeter, Appendix C records the research methodology, Appendix D gives sampled frequencies of identities of Section 5 and the norms of products of minimal vectors, and Appendix E tests the closure of other twists and determines the index of the span of the products.

All results were developed through a systematic human–AI collaboration, and the complete research record is public [repo]: every prompt, computation, and verification, end to end (Appendix C).

2 Preliminaries

2.1 The octonion algebra

Let $\mathbb{O}$ denote the real octonion algebra with basis $\{e_0, e_1, \ldots, e_7\}$, where e_0 is the identity. Its multiplication, the *standard octonion product*, is determined by the seven *Fano triples*

$$(1,2,4),\ (2,3,5),\ (3,4,6),\ (4,5,7),\ (5,6,1),\ (6,7,2),\ (7,1,3), \tag{2.1}$$

[1] The ambient algebra has no multiplicative identity (Remark 5.2), so the term "order" is used here in the nonunital sense.

encoding the rule: for each triple (a, b, c), $e_a \cdot e_b = +e_c$, with cyclic permutations carrying a positive sign and anti-cyclic permutations carrying a negative sign [Baez2002]. Each imaginary basis unit $e_1, \ldots, e_7$ squares to $-e_0$. Among *unital* real algebras, $\mathbb{O}$ is the unique normed division algebra of dimension 8 (Hurwitz); relaxing "unital normed division algebra" to the absence of zero divisors (equivalently, bijectivity of left and right multiplication by every nonzero element) admits further examples, including the nonunital Okubo algebra.

The *norm form* is $N(x) = x\bar{x} = \sum_{i=0}^{7} x_i^2$ for $x = \sum_i x_i e_i$, where $\bar{x} = x_0 e_0 - \sum_{i=1}^{7} x_i e_i$ is the conjugate. Throughout, *norm* means this squared Euclidean length, and $\langle x, y\rangle$ is the Euclidean inner product, so that $\langle x, x\rangle = N(x)$; on $\mathbb{R}^{24}$ both are summed over the three blocks. The composition property $N(xy) = N(x)N(y)$ holds. An octonion x is *purely imaginary* when $x_0 = 0$, that is, when $\bar{x} = -x$.

2.2 The E_8 lattice

In Wilson's description [Wilson2009], the E_8 lattice is the set $L \cong D_8^+$ of vectors in $\mathbb{R}^8$ whose coordinates (in the basis $\{e_0, \ldots, e_7\}$) are either all integers with even coordinate sum, or all half-integers with odd coordinate sum [ConwaySmith2003]. It has 240 roots (vectors of norm 2). Under the standard octonion product, L is closed: $L \cdot L \subseteq L$ (Lemma 4.2). At this normalization $e_0 \notin L$ (the lattice is even), so L is a nonunital order. As a lattice it is a $\sqrt{2}$-scaled copy of a maximal order of integral octonions, the unit-scale octavians, for which closure and maximality hold in the classical unital sense [Coxeter1946, ConwaySmith2003]. For a modern survey of the integral-octonion history, see [Petersson2018].

One maximal order of integral octonions is used throughout:

$$\mathcal{O} := \tfrac{1}{2}\,(e_1 + e_2)\,L. \tag{2.2}$$

The lattice $\mathcal{O}$ contains the eight basis units and is closed under the standard octonion product. It is one of the seven maximal orders of Section 1, the ring A_3 of Wilson [Wilson2009, Section 2], with Wilson's basis units 1 and i_t $(0 \le t \le 6)$ read as e_0 and e_{t+1}.[2] Since $(e_1 + e_2)^2 = -2$ and $L = -L$, the left alternative law $x(xy) = x^2 y$ gives $L = (e_1 + e_2)\,\mathcal{O}$.

2.3 Wilson's Leech lattice

Define $s = \frac{1}{2}(-e_0 + e_1 + \cdots + e_7)$ and $\bar{s} = \frac{1}{2}(-e_0 - e_1 - \cdots - e_7)$, both of norm $N(s) = N(\bar{s}) = 2$. Then $s \in L$ (it is a root in the half-integer branch, with an odd coordinate sum), while $\bar{s} \notin L$ (its coordinate sum is -4, which falls in the excluded even branch of the half-integer case). Let

$$L\bar{s} = \{\ell \cdot \bar{s} : \ell \in L\}, \qquad Ls = \{\ell \cdot s : \ell \in L\}. \tag{2.3}$$

Both $L\bar{s}$ and Ls are sublattices of L,[3] with $[L : L\bar{s}] = [L : Ls] = 16$, $L\bar{s} + Ls = L$, and $L\bar{s} \cap Ls = 2L$.

Definition 2.1 (Wilson)**.** The *Leech lattice* is

$$\Lambda = \big\{(x, y, z) \in L^3 : (1)\ x, y, z \in L;\ (2)\ x{+}y,\ x{+}z,\ y{+}z \in L\bar{s};\ (3)\ x{+}y{+}z \in Ls\big\}. \tag{2.4}$$

[2] [repo] `python_project/src/verify_structural_proof.py` [repo] `python_project/src/verify_sigma_sublattice_maximal_order.py`.

[3] $L\bar{s} \subseteq L$ even though $\bar{s} \notin L$: each $\ell \cdot \bar{s}$ with ℓ a root of L lies in L by direct computation, and the inclusion extends by $\mathbb{Z}$-linearity in the left factor. $L\bar{s}$ is therefore a $\mathbb{Z}$-subgroup of L. For Ls, the same follows from $s \in L$ and $L \cdot L \subseteq L$ (Lemma 4.2).

The 196,560 minimal vectors of Λ decompose into three families:

Family	Formula	Count
Type 1	$(2\lambda, 0, 0)$ and cyclic permutations	$3 \times 240 = 720$
Type 2	$(\lambda\bar{s},\ (\lambda\bar{s})j,\ 0)$ and cyclic permutations	$3 \times 240 \times 16 = 11{,}520$
Type 3	$((\lambda s)j,\ \lambda k,\ (\lambda j)k)$ and cyclic permutations	$3 \times 240 \times 16^2 = 184{,}320$

Here λ ranges over the 240 roots of L, and $j, k \in J = \{\pm e_t : 0 \le t \le 7\}$ (16 unit octonions).

3 The construction

Definition 3.1 (Transposition)**.** Let $\sigma = (i\ j)$ be a transposition on $\{1, \dots, 7\}$, extended to the linear involution $\sigma\colon \mathbb{R}^8 \to \mathbb{R}^8$ that fixes e_0 and swaps $e_i \leftrightarrow e_j$.

Definition 3.2 (σ-twisted octonion product)**.** The *σ-twisted octonion product* $\cdot_\sigma$ on $\mathbb{O}$ is defined from the standard octonion product (2.1) by

$$x \cdot_\sigma y \ :=\ \sigma\big(\sigma(x) \cdot \sigma(y)\big). \tag{3.1}$$

This is another octonion product on $\mathbb{O}$: by definition, $\sigma\colon (\mathbb{O}, \cdot) \to (\mathbb{O}, \cdot_\sigma)$ is an algebra isomorphism, $\sigma(x \cdot y) = \sigma(x) \cdot_\sigma \sigma(y)$.

Definition 3.3 (Triple product with $\mathbb{Z}_3$ routing)**.** Decompose $\mathbb{R}^{24} = \mathbb{O}_1 \oplus \mathbb{O}_2 \oplus \mathbb{O}_3$ into three copies of $\mathbb{R}^8$ (blocks indexed $\alpha = 1, 2, 3$). For $u = (x, y, z)$ and $v = (x', y', z')$ with $x, x' \in \mathbb{O}_1$, etc., define the *$\mathbb{Z}_3$-symmetric triple-octonion product*, or *triple product* for short,

$$u \star v = (P_1,\ P_2,\ P_3), \tag{3.2}$$

where each block P_α is the sum of three σ-twisted octonion products:

$$P_1 = x \cdot_\sigma x' + z \cdot_\sigma y' + y \cdot_\sigma z', \tag{3.3}$$

$$P_2 = y \cdot_\sigma y' + x \cdot_\sigma z' + z \cdot_\sigma x', \tag{3.4}$$

$$P_3 = z \cdot_\sigma z' + y \cdot_\sigma x' + x \cdot_\sigma y'. \tag{3.5}$$

The routing rule is: a product of blocks α and β lands in block γ, where $\{\alpha, \beta, \gamma\} = \{1, 2, 3\}$ for $\alpha \neq \beta$, and in block α when $\alpha = \beta$. All nine block-pair products use the same multiplication $\cdot_\sigma$.

4 Proof of closure

We prove Theorem 1.1 by showing that the σ-twisted octonion product preserves each of Wilson's three conditions (Definition 2.1). Throughout, and without loss of generality, $\sigma = (1\ 2)$.[4] The map σ fixes L setwise (Lemma 4.1) and carries the standard octonion product to an isomorphic one (Definition 3.2). Its load-bearing effect is that it *moves* the sublattices $L\bar{s}$ and Ls onto left multiples of the maximal order $\mathcal{O}$ of (2.2): $\sigma(L\bar{s}) = 2\mathcal{O}$ and $\sigma(Ls) = p\mathcal{O}$ for a purely imaginary $p \in \mathcal{O}$. Lemmas 4.3 and 4.4, the closure inclusions for the moved sublattices, then follow from the closure of $\mathcal{O}$.

[4] The 21 transpositions of imaginary basis units form a single orbit under conjugation by the order-21 group of coordinate permutations preserving the oriented Fano triples (2.1). Each such permutation is an octonion automorphism that fixes s and $\bar{s}$ and preserves L, so conjugating by it carries the proof below for (1 2) to any other transposition.

4.1 The four lemmas

Lemma 4.1. $\sigma(L) = L$.

Proof. The transposition σ permutes two coordinates, so it preserves the two conditions that define L (Section 2): all coordinates integers or all half-integers, and the parity of the coordinate sum. Hence $\sigma(L) \subseteq L$. Since σ is an involution, $\sigma(L) = L$. □

The remaining three lemmas rest on the maximal order $\mathcal{O}$ of (2.2) and on three facts. First, two lattice identities: with $p := e_2 - e_4 + e_5 - e_6$, a purely imaginary element of $\mathcal{O}$ of norm 4,

$$\sigma(L\bar{s}) = 2\mathcal{O}, \qquad \sigma(Ls) = p\mathcal{O}. \tag{4.1}$$

The inclusions $2\mathcal{O} \subseteq \sigma(L\bar{s})$ and $p\mathcal{O} \subseteq \sigma(Ls)$ are checked on a $\mathbb{Z}$-basis $\{b_k\}$ of $\mathcal{O}$: the eight vectors $2b_k$ lie in $\sigma(L\bar{s})$, and the eight products $p\,b_k$ lie in $\sigma(Ls)$. Both sides of each identity have covolume 16. Left multiplication by x scales inner products by $N(x)$ and hence covolumes by $N(x)^4$. Since $L = (e_1 + e_2)\,\mathcal{O}$ has covolume 1 (E_8 is unimodular), $\mathcal{O}$ has covolume $\frac{1}{16}$, so $2\mathcal{O}$ and $p\mathcal{O}$ have covolume 16. So do $\sigma(L\bar{s})$ and $\sigma(Ls)$, by $[L : L\bar{s}] = [L : Ls] = 16$ (Section 2.3). A sublattice of the same covolume is the whole lattice.[5]

Second, an identity in the octonions: for purely imaginary x and all $a, b \in \mathbb{O}$,

$$(xa)(xb) = x\big((ax)b\big). \tag{4.2}$$

Indeed, both sides vanish for $x = 0$. For $x \neq 0$ we have $x^2 = -x\bar{x} = -N(x)$, so $a = yx$ with $y := -ax/N(x)$ by the right alternative law $(ax)x = ax^2$, and $xa = xyx$ by the flexible law $x(yx) = (xy)x$. The left Moufang identity $(xyx)z = x(y(xz))$ [ConwaySmith2003] with $z = xb$ and the left alternative law $x(xb) = x^2 b$ give $(xa)(xb) = x\big(y(x^2 b)\big) = -N(x)\,x(yb) = x\big((ax)b\big)$.

Third, its consequence for lattices: if $M \subseteq \mathbb{O}$ is closed under the standard octonion product and $x \in M$ is purely imaginary, then by (4.2)

$$(xM)(xM) \subseteq x\big((Mx)M\big) \subseteq xM. \tag{4.3}$$

Lemma 4.2. $L \cdot L \subseteq L$.

Proof. $L = (e_1 + e_2)\,\mathcal{O}$ by (2.2), with $e_1 + e_2$ a purely imaginary element of $\mathcal{O}$; apply (4.3) with $M = \mathcal{O}$. This is the norm-2-scale form of the classical fact that the maximal orders of integral octonions are closed under multiplication [Coxeter1946, ConwaySmith2003]. □

Lemma 4.3. $L \cdot \sigma(L\bar{s}) \subseteq \sigma(L\bar{s})$.

Proof. $L = (e_1 + e_2)\,\mathcal{O} \subseteq \mathcal{O} \cdot \mathcal{O} \subseteq \mathcal{O}$, so by (4.1) $L \cdot \sigma(L\bar{s}) = L \cdot 2\mathcal{O} \subseteq 2\,(\mathcal{O} \cdot \mathcal{O}) \subseteq 2\mathcal{O} = \sigma(L\bar{s})$. □

Lemma 4.4. $\sigma(Ls) \cdot \sigma(Ls) \subseteq \sigma(Ls)$.

Proof. $\sigma(Ls) = p\mathcal{O}$ by (4.1), with p a purely imaginary element of $\mathcal{O}$; apply (4.3) with $M = \mathcal{O}$. □

Remark 4.5. The sublattice Ls is *not* closed under the standard octonion product: $u_1 = (-1, 0, 0, 0, 1, 0, 1, 1)$ and $u_2 = (-1, 1, 0, 0, 0, 1, 0, 1)$ lie in Ls; $u_1 \cdot u_2 = (0, -2, 2, 1, -2, -1, -1, -1)$ lies in $L\bar{s} \setminus Ls$. Moreover $\sigma(Ls) \neq Ls$: with $\sigma = (1\ 2)$, the element $v = (-1, 0, 1, 0, 0, 1, 0, 1)$ lies

[5] [repo] `python_project/src/verify_structural_proof.py`.

in $\sigma(Ls) \setminus Ls$. Likewise $\sigma(L\bar{s}) \neq L\bar{s}$: $w = (0,0,-1,0,-1,0,-1,-1) \in \sigma(L\bar{s}) \setminus L\bar{s}$. The closure of $\sigma(Ls)$ is not inherited from an ideal property: $\sigma(Ls) = p\mathcal{O}$ is not an ideal of L on either side, since neither $L \cdot \sigma(Ls)$ nor $\sigma(Ls) \cdot L$ lies in $\sigma(Ls)$.[6] By contrast, $\sigma(L\bar{s}) = 2\mathcal{O}$ is a two-sided ideal of L, of which Lemma 4.3 is the left-handed half.

4.2 Blockwise form of the triple product

Since each σ-twisted product has the form $a \cdot_\sigma b = \sigma(\sigma(a) \cdot \sigma(b))$ and σ is linear, each block P_α is σ applied to a sum of standard octonion products of σ-images of the inputs. Write $X = \sigma(x)$, $Y = \sigma(y)$, $Z = \sigma(z)$, $X' = \sigma(x')$, $Y' = \sigma(y')$, $Z' = \sigma(z')$:

$$P_1 = \sigma\big(X \cdot X' + Z \cdot Y' + Y \cdot Z'\big), \tag{4.4}$$

$$P_2 = \sigma\big(Y \cdot Y' + X \cdot Z' + Z \cdot X'\big), \tag{4.5}$$

$$P_3 = \sigma\big(Z \cdot Z' + Y \cdot X' + X \cdot Y'\big). \tag{4.6}$$

Taking sums:

- **Block sum.** $P_1 + P_2 + P_3 = \sigma\big((X+Y+Z) \cdot (X'+Y'+Z')\big)$.
- **Pairwise sums.** For example, $P_1 + P_2 = \sigma\big(X \cdot (X'+Z') + Y \cdot (Y'+Z') + Z \cdot (X'+Y')\big)$, and similarly for the other two pairs.

4.3 Proof of Wilson's three conditions

4.3.1 Condition (1): each block lies in L

Proposition 4.6. *For all $u, v \in \Lambda$, each block P_α of $u \star v$ lies in L.*

Proof. Consider $P_1 = \sigma(X \cdot X' + Z \cdot Y' + Y \cdot Z')$. Since $u \in \Lambda$, condition (1) gives $x \in L$, so $X = \sigma(x) \in L$ by Lemma 4.1. Similarly $X', Y, Y', Z, Z' \in L$. By Lemma 4.2, each product $X \cdot X'$, $Z \cdot Y'$, $Y \cdot Z'$ lies in L. Their sum lies in L (closed under addition). Applying σ and using Lemma 4.1, $P_1 \in L$. The same argument applies to P_2 and P_3. □

4.3.2 Condition (2): pairwise sums lie in $L\bar{s}$

Proposition 4.7. *For all $u, v \in \Lambda$, the pairwise sums $P_\alpha + P_\beta$ lie in $L\bar{s}$.*

Proof. We show $P_1 + P_2 \in L\bar{s}$; the other two cases follow by the $\mathbb{Z}_3$ symmetry of the routing.

From (4.4)–(4.5):

$$P_1 + P_2 = \sigma\big(X \cdot (X'+Z') + Y \cdot (Y'+Z') + Z \cdot (X'+Y')\big). \tag{4.7}$$

Since $v \in \Lambda$, condition (2) gives $x' + z' \in L\bar{s}$, so $X' + Z' = \sigma(x' + z') \in \sigma(L\bar{s})$. Similarly $Y' + Z' \in \sigma(L\bar{s})$ and $X' + Y' \in \sigma(L\bar{s})$. Since $u \in \Lambda$, we have $X, Y, Z \in L$ (by Lemma 4.1).

By Lemma 4.3, each product $X \cdot (X'+Z')$, $Y \cdot (Y'+Z')$, $Z \cdot (X'+Y')$ lies in $\sigma(L\bar{s})$. Their sum lies in $\sigma(L\bar{s})$. Applying the outer σ:

$$P_1 + P_2 \in \sigma(\sigma(L\bar{s})) = L\bar{s}, \tag{4.8}$$

since $\sigma^2 = \mathrm{id}$. □

[6] [repo] `python_project/src/verify_sigma_Ls_ideal_exclusion.py`.

4.3.3 Condition (3): the block sum lies in Ls

This is the crux of the proof. With the standard octonion product in place of $\cdot_\sigma$, the block sum would require $Ls \cdot Ls \subseteq Ls$, which does not hold (Remark 4.5). The twist redirects the computation through $\sigma(Ls)$, where closure holds.

Proposition 4.8. *For all $u, v \in \Lambda$, the block sum $P_1 + P_2 + P_3$ lies in Ls.*

Proof. From (4.4)–(4.6):

$$P_1 + P_2 + P_3 = \sigma\big((X + Y + Z) \cdot (X' + Y' + Z')\big). \tag{4.9}$$

Since $u \in \Lambda$, condition (3) gives $x + y + z \in Ls$, so

$$X + Y + Z = \sigma(x + y + z) \in \sigma(Ls). \tag{4.10}$$

Similarly $X' + Y' + Z' \in \sigma(Ls)$.

By Lemma 4.4, $\sigma(Ls) \cdot \sigma(Ls) \subseteq \sigma(Ls)$, so

$$(X + Y + Z) \cdot (X' + Y' + Z') \in \sigma(Ls). \tag{4.11}$$

Applying the outer σ:

$$P_1 + P_2 + P_3 \in \sigma(\sigma(Ls)) = Ls. \tag{4.12}$$

□

This completes the proof of Theorem 1.1.

5 Algebraic properties

Throughout this section, $\sigma = (1\ 2)$, as in Section 4. Appendix E complements this section: it tests other twists (Section 5.6) for closure on Λ and determines how much of Λ the products of the order reach.

5.1 Identities

Several standard algebraic identities, among them associativity, alternativity, flexibility, power-associativity, and commutativity, fail on the ambient algebra $(\mathbb{R}^{24}, +, \star)$ and hence, by scaling, on the order $(\Lambda, +, \star)$. The table below lists a witness of each violation, which is easily checked by hand from Definitions 3.2 and 3.3.[7] Appendix D estimates by sampling how often the identities from left alternativity to symmetric composition hold on minimal vectors.

[7] [repo] `python_project/src/verify_identity_witnesses.py`.

Identity	Witness of violation $(u, v\,[, w])$
Associativity $(u \star v) \star w = u \star (v \star w)$	$(e_0,0,0)$, $(0,e_0,0)$, $(0,0,e_0)$
Left alternativity $(u \star u) \star v = u \star (u \star v)$	$(e_0,0,0)$, $(0,e_0,0)$
Right alternativity $(u \star v) \star v = u \star (v \star v)$	$(0,e_0,0)$, $(e_0,0,0)$
Flexibility $(u \star v) \star u = u \star (v \star u)$	$(e_3,e_4,0)$, $(e_0,0,0)$
Cube power-associativity $(u \star u) \star u = u \star (u \star u)$	(e_0,e_3,e_4)
Quartic power-associativity: the five parenthesizations of $u \star u \star u \star u$ coincide	$(e_0,-e_0,0)$
Commutativity $u \star v = v \star u$	$(e_3,0,0)$, $(e_4,0,0)$
Norm multiplicativity $N(u \star v) = N(u)N(v)$	(e_0,e_0,e_0), $(e_0,-e_0,0)$
Symmetric composition $\langle u \star v, w\rangle = \langle u, v \star w\rangle$	$(e_0,0,0)$, $(e_3,0,0)$, $(e_3,0,0)$
Multiplicative identity	no identity element exists (Remark 5.2)

The failure of norm multiplicativity rules out $(\Lambda, +, \star)$ being a composition algebra for N. The failure of symmetric composition rules out, independently, its being a symmetric composition algebra for N, such as the Okubo and para-Hurwitz algebras (Alberto Elduque [Elduque2000, Elduque2023]; Susumu Okubo [Okubo1995]).

Remark 5.1. The product does not preserve the minimal shell: for the minimal vector $u = (2\lambda, 0, 0)$ with $\lambda = e_3 + e_4$ (Type 1 in Section 2.3), $u \star u = (4\lambda^2, 0, 0) = (-8e_0, 0, 0)$, of norm 64. Appendix D records the self-product norms of all minimal vectors and the pair-product norms on sampled pairs.

Remark 5.2. The ambient algebra $(\mathbb{R}^{24}, +, \star)$ has no multiplicative identity. A left identity (a, b, c) would act on (x', y', z') by the block matrix $\left(\begin{smallmatrix} a & c & b \\ c & b & a \\ b & a & c \end{smallmatrix}\right)$, with entries acting by left multiplication in $(\mathbb{O}, \cdot_\sigma)$; each of a, b, c occurs both on and off the diagonal, so it would have to act both as the identity and as zero. In particular, (e_0, e_0, e_0) is not an identity: it maps (x', y', z') to $(x' + y' + z',\ x' + y' + z',\ x' + y' + z')$. Consequently $(\Lambda, +, \star)$ is a nonunital order.

The defining property of the order is *closure*: the Leech lattice is stable under the product.

5.2 Idempotents

The ambient algebra $(\mathbb{R}^{24}, +, \star)$ has exactly eight idempotents, by symbolic classification.[8] Writing $\varepsilon_1 = (e_0, 0, 0)$, $\varepsilon_2 = (0, e_0, 0)$, $\varepsilon_3 = (0, 0, e_0)$, and $\omega = \varepsilon_1 + \varepsilon_2 + \varepsilon_3$, they are

$$0, \qquad \varepsilon_1,\ \varepsilon_2,\ \varepsilon_3, \qquad \tfrac{1}{3}\omega, \qquad \varepsilon_i - \tfrac{1}{3}\omega \ \ (i = 1, 2, 3). \tag{5.1}$$

Note that every block of every one of them is a multiple of e_0.

The order $(\Lambda, +, \star)$ has no nonzero idempotent:[9] each nonzero idempotent spans a rational ray that meets Λ, and its least positive multiple in Λ is

$$4\varepsilon_i \ (N = 16), \qquad 4\omega \ (N = 48), \qquad 6\big(\varepsilon_i - \tfrac{1}{3}\omega\big) \ (N = 24). \tag{5.2}$$

These multiples follow from Wilson's three conditions. Since L, $L\bar{s}$, and Ls meet $\mathbb{R}e_0$ in $2\mathbb{Z}e_0$, $2\mathbb{Z}e_0$, and $4\mathbb{Z}e_0$, a real multiple of an idempotent lies in Λ exactly when its blocks lie in $2\mathbb{Z}e_0$

[8] [repo] `python_project/src/verify_idempotent_classification.py`. The nonzero idempotents of (5.1) are also obtained in [Petersson2026].

[9] [repo] `python_project/src/verify_idempotent_lattice_rescaling.py`.

(condition (1), which then implies condition (2)) and its block sum lies in $4\mathbb{Z}e_0$ (condition (3)). The intersections with $L\bar{s}$ and Ls follow from $2e_0 \in \sigma(2\mathcal{O}) = L\bar{s}$ (by (4.1), as σ fixes $e_0 \in \mathcal{O}$) and from $L\bar{s} \cap Ls = 2L$, which contains $4e_0$ but not $2e_0$. On the line spanned by a nonzero idempotent, the points of Λ are the integer multiples kw of the corresponding vector w in (5.2), of norm $16k^2$, $48k^2$, or $24k^2$; in particular, no real multiple of an idempotent lies on the minimal shell.

5.3 Maximality

The order $(\Lambda, +, \star)$ is not maximal: larger orders in the ambient algebra contain it. Two such orders, L^3 and $\Lambda + \mathbb{Z}\varepsilon_1$, witness this, and neither contains the other.

The first witness is L^3. The proof of Proposition 4.6 uses only Wilson's condition (1), that the blocks of both factors lie in L, so it shows $L^3 \star L^3 \subseteq L^3$. The lattice L^3 is even and integral, and $[L^3 : \Lambda] = 2^{12}$.

The second witness is $\Lambda + \mathbb{Z}\varepsilon_1$, the lattice generated by Λ and the idempotent $\varepsilon_1 = (e_0, 0, 0)$. On either side, ε_1 acts as the swap of blocks 2 and 3: since e_0 is the identity of $(\mathbb{O}, \cdot_\sigma)$, (3.3)–(3.5) give $\varepsilon_1 \star v = v \star \varepsilon_1 = (x', z', y')$ for $v = (x', y', z')$. This swap preserves Λ, because Wilson's three conditions are symmetric in the blocks. With $\varepsilon_1 \star \varepsilon_1 = \varepsilon_1$, the lattice $\Lambda + \mathbb{Z}\varepsilon_1$ is therefore $\star$-closed by bilinearity. By (5.2), $4\varepsilon_1$ is the least positive multiple of ε_1 in Λ, so $[\Lambda + \mathbb{Z}\varepsilon_1 : \Lambda] = 4$. Unlike L^3, the second witness is not integral: ε_1 has half-integer inner products with some vectors of Λ.

Neither witness contains the other: $\Lambda + \mathbb{Z}\varepsilon_1$ is not contained in L^3, because the first block e_0 of ε_1 is not in L, and L^3 is not contained in $\Lambda + \mathbb{Z}\varepsilon_1$, since $[L^3 : \Lambda] = 2^{12}$ exceeds $[\Lambda + \mathbb{Z}\varepsilon_1 : \Lambda] = 4$.

5.4 The automorphism group

Restricted to the Leech lattice, the ambient algebra has the automorphism group $\mathrm{Aut}(\Lambda, +, \star)$, the group of bijections of Λ that preserve both $+$ and $\star$. This group is finite, of order 36, with structure $C_6 \times S_3$. The S_3 permutes the three blocks of Definition 3.3, and the C_6 is generated by the following signed permutation of imaginary basis units, an automorphism of the algebra $(\mathbb{O}, \cdot_\sigma)$ of Definition 3.2, applied in each block, with e_0 fixed:

$$e_1 \mapsto -e_5, \quad e_5 \mapsto e_6, \quad e_6 \mapsto e_1, \tag{5.3}$$

$$e_2 \mapsto -e_7, \quad e_7 \mapsto -e_3, \quad e_3 \mapsto e_2, \tag{5.4}$$

$$e_4 \mapsto -e_4. \tag{5.5}$$

Every automorphism of the order $(\Lambda, +, \star)$ is an isometry of Λ, so $\mathrm{Aut}(\Lambda, +, \star)$ is a subgroup of the Conway group Co_0, the isometry group of Λ. This inclusion can already be settled at the level of the ambient algebra, to which every automorphism of the order extends linearly (Λ spans $\mathbb{R}^{24}$, and $\star$ is bilinear). As a real algebra, $\mathbb{R}^{24} = \mathbb{O}^3$ splits as $D \oplus T$, where $D = \{(a, a, a) : a \in \mathbb{O}\}$ (of dimension 8) and $T = \{(x, y, z) \in \mathbb{O}^3 : x + y + z = 0\}$ (of dimension 16). Both are two-sided ideals, and $D \star T = T \star D = 0$. Every automorphism of the ambient algebra preserves them and acts on each by an octonion automorphism, composed with a permutation of the blocks, so that the ambient algebra itself has the automorphism group $\mathrm{Aut}(\mathbb{R}^{24}, \star) = G_2 \times G_2 \times S_3$, with G_2 the automorphism group of the octonions.[10] An automorphism of the *real* octonions preserves the positive-definite octonion norm, and $D \perp T$; hence every automorphism of the ambient algebra is orthogonal, and

$$\mathrm{Aut}(\Lambda, +, \star) \subsetneq \mathrm{Co}_0\,. \tag{5.6}$$

[10] [Petersson2026].

The inclusion is strict because $-I_{24} \in \mathrm{Co}_0$ does not preserve $\star$.

The derivation, the complete enumerations, and an independent verification are recorded separately in a companion note.[11]

5.5 Square-zero elements

In the ambient algebra, $u = (x, y, z)$ satisfies $u \star u = 0$ if and only if x, y, z are purely imaginary, of equal norm, and sum to zero (again by symbolic classification). Three vectors of equal norm summing to zero lie at 120° to one another; with their negatives they are the six vertices of a regular hexagon, and x, y, z are three alternating vertices, a *hexagonal triple*. The solution set is a cone of dimension 12, and it lies in T.[12]

The lattice does contain such elements. A vector $(x, y, z) \in \Lambda$ squares to zero exactly when x, y, z are purely imaginary vectors of $L\bar{s}$, have equal norms, and sum to zero (for $x + y + z = 0$, Wilson's three conditions reduce to $x, y, z \in L\bar{s}$). The 126 purely imaginary vectors of norm 4 in $L\bar{s}$ form an E_7 root system: by (4.1), $L\bar{s} = \sigma(2\mathcal{O})$ is an E_8 lattice whose minimal vectors have norm 4 and include $2e_0$, and these are its minimal vectors orthogonal to $2e_0$. Each of these roots has exactly 32 roots at 120°; the third block is then forced, so the ordered hexagonal triples number $126 \times 32 = 4{,}032$, one square-zero vector of Λ each, all of norm 12. The norm of every square-zero vector of Λ lies in $12\mathbb{Z}$, since $N(u) = 3N(x)$ and every norm of $L\bar{s}$ lies in $4\mathbb{Z}$. These 4,032 vectors are exactly the square-zero vectors of norm 12, and norm 24 occurs as well: the sum of two of them that are blockwise orthogonal is square-zero of norm 24.[13]

5.6 The closure criterion

The construction twists the octonion product by one transposition, σ. This subsection replaces σ by other maps and characterizes those for which the resulting triple product still closes on Λ.

A *twist* is an invertible linear map g of $\mathbb{R}^8$ with $g(L) = L$. It defines the twisted product $x \cdot_g y := g^{-1}\big(g(x) \cdot g(y)\big)$ on $\mathbb{O}$. The triple product $\star_g$ is that of Definition 3.3, with $\cdot_g$ in all nine block-pair products. Since $\sigma^{-1} = \sigma$, the twist $g = \sigma$ (Lemma 4.1) gives the product of Definition 3.2, and $\star_\sigma = \star$.

Proposition 5.3. *For every twist g, the product $\star_g$ closes on Λ if and only if $L\bar{s}$ and Ls are closed under $\cdot_g$, that is, if and only if $g(L\bar{s})$ and $g(Ls)$ are closed under the standard octonion product.*

Proof. Write xy for $x \cdot_g y$, while $L\bar{s}$ and Ls remain the lattices of (2.3). Since $g(L) = L$, Lemma 4.2 gives $L \cdot_g L = g^{-1}(L \cdot L) \subseteq L$, so every block of a product of vectors of L^3 lies in L. For $u = (x, y, z) \in L^3$ put $a = x + y$, $b = x + z$, and $w = x + y + z$. As $y + z = a + b - 2x$ and $2L \subseteq L\bar{s}$ (Section 2.3), $u \in \Lambda$ exactly when $a, b \in L\bar{s}$ and $w \in Ls$, and then $u = (a + b - w,\ w - b,\ w - a)$. For two such vectors u and u', Definition 3.3 gives

$$P_1 + P_2 + P_3 = ww', \qquad P_1 + P_2 = ab' + ba' - aa' + 2(bb' - wb' + ww' - bw'),$$

and the other two pairwise sums follow by the $\mathbb{Z}_3$ symmetry of the routing. If $L\bar{s}$ and Ls are closed, the block sum lies in Ls and, since $2L \subseteq L\bar{s}$, the pairwise sums lie in $L\bar{s}$. Conversely, $(a, 0, -a)$

[11] [repo] `paper/automorphism_group_2026-07-12.pdf`; [repo] `python_project/src/verify_aut_lambda_star.py` [repo] `gap_project/aut_lambda_star.g`.

[12] [repo] `python_project/src/verify_square_zero_classification.py`.

[13] [repo] `python_project/src/verify_square_zero_norm24_witness.py`.

and $(w, w, -w)$ lie in Λ for $a \in L\bar{s}$ and $w \in Ls$; the product of two vectors of the first kind has $P_1 + P_2 = -aa'$, and that of two of the second kind has block sum ww', so closure forces both inclusions. □

For $g = \sigma$, the lattices $\sigma(L\bar{s}) = 2\mathcal{O}$ and $\sigma(Ls) = p\mathcal{O}$ of (4.1) are closed under the standard octonion product, by the closure of $\mathcal{O}$ and Lemma 4.4, so the proposition gives Theorem 1.1 again.

6 Related work

Wilson (2009) [Wilson2009] characterizes Λ as $\{(x, y, z) \in L^3 : \text{conditions (1)–(3)}\}$, gives an explicit listing of the 196,560 minimal vectors in the form $3 \times 240 \times (1 + 16 + 16 \times 16)$, and proves that Λ is even and self-dual in his halved-norm convention. The sublattice description is the setting used in the present paper.

Dixon (2010) [Dixon2010] obtains the integral octonions from the octonion XY-product

$$A \circ_{XY} B \;=\; (AX)(Y^{\dagger}B), \tag{6.1}$$

where Dixon writes $X^{\dagger}$ for the conjugate $\bar{X}$, and $X, Y \in \mathbb{O}$ are chosen so that $YX^{\dagger}$ is a unit octonion (the identity of the new multiplication). When $Y^{\dagger} = X^{-1}$ the XY-product specializes to the X-product $A \circ_X B = (AX)(X^{-1}B)$; Dixon uses this X-product with $X = \ell_0^{-1}$, where $\ell_0 = \frac{1}{2}(e_0 + e_1 + \cdots + e_7)$, to obtain E_8, and XY-products for the laminated lattice Λ_{16}. For Λ, Dixon writes the minimal vectors of a copy of Λ (the componentwise conjugates of Wilson's [Wilson2009]) in $\mathbb{O}^3$ with the standard octonion product.[14] We have independently reproduced this construction.[15]

Baez–Egan (2014). In part 9 of the series *Integral octonions* [Baez2014], Greg Egan constructs the Leech lattice from the E_8 lattice alone; in the form of part 11, for sublattices $L_1, L_2 \subset L$, each isometric to $\sqrt{2}\,L$, with $L_1 \cap L_2 = 2L$, the lattice

$$\Lambda(L_1, L_2) \;:=\; \big\{(x, y, z) \in L^3 : x{+}y,\; x{+}z,\; y{+}z \in L_1;\; x{+}y{+}z \in L_2\big\} \tag{6.2}$$

is isometric to Λ. Wilson's lattice is $\Lambda = \Lambda(L\bar{s}, Ls)$ (Definition 2.1), and, with σ applied in each block, $\sigma(\Lambda) = \Lambda(2\mathcal{O}, p\mathcal{O})$ by (4.1) and Lemma 4.1. Applied in each block in the same way, an isometry g of L with $g(L\bar{s}) = L_1$ and $g(Ls) = L_2$, which exists by Appendix A, carries Λ onto $\Lambda(L_1, L_2)$ and $\star_g$ onto the untwisted product $\star_{\mathrm{id}}$ of Section 5.6. By Proposition 5.3, then, $\star_{\mathrm{id}}$ closes on $\Lambda(L_1, L_2)$ exactly when L_1 and L_2 are closed under the standard octonion product: it closes on $\sigma(\Lambda)$, which is Theorem 1.1, and not on Λ (Remark 4.5). There are 270 sublattices of L isometric to $\sqrt{2}\,L$, and each meets exactly 64 of them in $2L$, so Egan's construction gives $17{,}280 = 270 \times 64$ lattices, all distinct (part 11, and Egan's comments there [Egan2014]). Appendix A reads these pairs modulo 2, with the product that $L/2L$ inherits from the standard octonion product.

In parts 9 and 10, Baez considers the 27-dimensional lattice $\mathbb{Z}^3 \oplus L_L \subset \mathfrak{h}_3(\mathbb{O})$ of the exceptional Jordan algebra, with integers on the diagonal and an off-diagonal Leech lattice L_L at the standard scale, with entries in a maximal order of integral octonions (here $\mathcal{O}$), so that $L_L \cong \Lambda/\sqrt{2}$. Egan found [Egan2014] that this lattice is *not* closed under the Jordan product $x \circ y = xy + yx$ but *is*

[14] The construction has an earlier precursor in Dixon [Dixon1995], where a closely related octonionic description of Λ is proposed. A 2005 preprint of Dixon's [Dixon2005] carried a formula whose third family of minimal vectors required a small correction, which Wilson [Wilson2009] supplied in a bracketed note while independently deriving the sublattice characterization. The situation is similar to the Kirmse–Bruck–Coxeter relationship recorded in Appendix B: an early, structurally correct proposal needing correction.

[15] [repo] `python_project/src/leech_dixon.py` [repo] `python_project/src/e8_dixon.py`.

closed under the doubled product $2(xy+yx)$. Egan further enumerates at least 17,280 distinct such lattices L_L for which the associated 27-dimensional lattice is closed under this doubled product: those of Egan's construction with $\mathcal{O}$ in place of L (see part 10 for Egan's computational counts). Baez leaves open the question whether the 27-dimensional lattice of any such L_L is closed under the undoubled Jordan product, and conjectures that the answer is negative. We have likewise independently reproduced the full Baez–Egan verification chain: Egan's simple roots of E_8 and the two sublattices built from them, the lemma that these meet in twice that E_8 lattice, the Leech-lattice theorem for that pair, the doubled-Jordan-product closure,[16] the derivation of the lower bound 244,035,421 on the number of sublattices of L^3 isometric to Λ,[17] and Egan's count.[18]

Anatomy of the Baez–Egan closure on Λ. The Baez–Egan closure is on the 27-dimensional lattice $\mathbb{Z}^3 \oplus L_L$, not on L_L alone. At the paper's scale, for off-diagonal Hermitian matrices $M_u, M_v \in \mathfrak{h}_3(\mathbb{O})$ representing $u = (x, y, z)$ and $v = (x', y', z')$ in $\Lambda \subset \mathbb{O}^3$, direct computation gives

$$2(M_u M_v + M_v M_u) \;=\; \Delta \;+\; R, \tag{6.3}$$

where R is an off-diagonal residue lying in $(2L)^3 \subseteq \Lambda$ and $\Delta \in \mathbb{Z}^3$ is a diagonal piece whose entries are four times the block-pair Euclidean inner products of u and v:

$$\Delta \;=\; 4\big(\langle x, x'\rangle + \langle y, y'\rangle,\; \langle x, x'\rangle + \langle z, z'\rangle,\; \langle y, y'\rangle + \langle z, z'\rangle\big), \tag{6.4}$$

so $\operatorname{tr}\Delta = 8\langle u, v\rangle$ recovers the $\mathbb{R}^{24}$ inner product of u and v. Projecting onto the off-diagonal subspace defines a bilinear product $\varphi(u, v) := R$ under which Λ is closed, so that $(\Lambda, +, \varphi)$ is an order in the sense of Corollary 1.2. For minimal vectors $u \in \operatorname{Min}(\Lambda)$, squaring gives $\operatorname{tr}\Delta = 8N(u) = 64$, so the projection is nontrivial on every minimal-vector input. The induced product φ is moreover commutative (the Jordan product is symmetric in its arguments), whereas $\star$ is not (Section 5.1). The products φ and $\star$ are therefore distinct bilinear products on $\mathbb{R}^{24}$ that both close on Λ. Their spans differ as well. Computed by the same Hermite-normal-form method as for $\star$ (Appendix E),[19] the span index of φ is 2^{25}. Unlike for $\star$, the span of φ does not contain 2Λ: its index exceeds $[\Lambda : 2\Lambda] = 2^{24}$.

Earlier precursors. Three works of 1982 and 1996 construct Λ from three copies of E_8 or relate it to the exceptional Jordan algebra. John H. Conway and Neil J. A. Sloane [ConwaySloane1982] give 23 constructions of Λ, one for each Niemeier lattice other than the Leech lattice, including a construction from three copies of E_8. James Lepowsky and Arne Meurman [LepowskyMeurman1982] describe Λ as a sublattice of E_8^3 containing $2E_8^3$ and generate the Conway group Co_0 from a maximal 2-local subgroup, which permutes the three E_8 factors, and one extra automorphism. Noam D. Elkies and Benedict H. Gross [ElkiesGross1996] relate an integer form of the exceptional Jordan algebra $\mathfrak{h}_3(\mathbb{O})$, with a positive definite element of determinant 1 outside the orbit of the identity matrix, to Λ and to the even lattice of rank 26 and determinant 3 with no vectors of squared length 2.

Leonard E. Dickson (1923), Johannes Kirmse (1924), Kurt Mahler (1942), Richard H. Bruck and H. S. M. Coxeter (1946). For the history, see Appendix B.

Daniele Corradetti (2026). Corradetti [Corradetti2026] studies integral structures on E_8 through the Okubo algebra, introducing an integral order-3 automorphism of the octonions whose

[16] [repo] `python_project/src/verify_egan_construction.py`.
[17] An orbit-stabilizer bound, $\lceil 3!\,|W(E_8)|^3/|\mathrm{Co}_0|\rceil$, with $W(E_8)$ the Weyl group of E_8; [repo] `python_project/src/verify_egan_lower_bound.py`.
[18] [repo] `python_project/src/egan_baez_count.py`.
[19] [repo] `python_project/src/verify_phi_span_index.py`.

Petersson isotope is the compact real Okubo algebra. The Coxeter–Dickson E_8 lattice then carries an octonionic, a para-octonionic, and an Okubo integral structure, three multiplicative systems that are not isomorphic.

7 Conclusion

The Leech lattice Λ is an order under the $\mathbb{Z}_3$-symmetric triple-octonion product of Definition 3.3 (Corollary 1.2), in Wilson's sublattice description. The proof (Section 4) rests on one maximal order $\mathcal{O}$ of integral octonions: the transposition σ fixes L and moves Wilson's sublattices $L\bar{s}$ and Ls onto the left multiples $2\mathcal{O}$ and $p\mathcal{O}$, and the closure of $\mathcal{O}$ gives the inclusions the proof needs. More generally, for every twist g, the product $\star_g$ closes on Λ exactly when $L\bar{s}$ and Ls are closed under $\cdot_g$ (Proposition 5.3). The order is nonunital, has no nonzero idempotent, and is not maximal; its automorphism group is $C_6 \times S_3$, and its span index is 2^{16}, a value that depends on the twist (Section 5 and Appendix E).

Several questions remain open:

- Where $\mathrm{Aut}(\Lambda, +, \star)$ sits inside Co_0 up to conjugacy, and whether the order-6 automorphism of $(\mathbb{O}, \cdot_\sigma)$ generating the C_6 (Section 5.4) admits an intrinsic description.

- A classification of the algebra $(\mathbb{R}^{24}, +, \star)$, which is nonassociative, nonalternative, nonflexible, non-power-associative, noncommutative, and nonunital (Section 5.1).

- A reason for the closure pattern of Appendix E beyond the 2-cycles, in particular for the failure of every 4-cycle and for the exact one-half rates.

8 Outlook

A more natural classification of $(\mathbb{R}^{24}, +, \star)$ may emerge from a *ternary* rather than binary viewpoint. The Leech lattice carries a prominent $\mathbb{Z}_3$ symmetry (the cyclic permutation of the three octonion blocks), and $\star$ is assembled from nine bilinear building blocks routed by that $\mathbb{Z}_3$ action. Symmetric and ternary composition algebras, as studied by Elduque [Elduque1996, Elduque2000, Elduque2023] and others [SmithVojtechovsky2022, KamiyaOkubo2015, Okubo1995], are typically nonassociative, noncommutative, nonalternative, and non-power-associative, the same negative classification $(\mathbb{R}^{24}, +, \star)$ satisfies. Yet they carry high internal symmetry, with order-3 automorphisms playing the central role: the Okubo algebra has automorphism group $\mathrm{SU}(3)/\mathbb{Z}_3$, and its product is built from an order-3 automorphism of the octonions. Jonathan I. Hall's monograph [Hall2019] establishes that Moufang loops and groups with triality are essentially the same object. Recasting $\star$ as a trace-like reduction of a ternary operation compatible with the $\mathbb{Z}_3$-block symmetry is a natural direction for further work.

Programs that model fundamental physics on normed-division-algebra, triality, S_3 symmetry, or symmetric-composition-algebra structures [GresnigtGourlayVarma2023, Koeplinger2023, FureyHughes2025, MarraniCorradettiZucconi2025] are one setting in which a nonclassical bilinear product on $\mathbb{R}^{24}$ may be of interest.

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Declarations

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Data and code availability. All data and code are public: the complete research record, including every verification script cited in this paper, is available at the repositories of [repo].

Use of AI. AI assistance is a stated, material part of this work, including the drafting of the manuscript text; the methodology, the role of AI, and the human-designed verification and governance process are documented in Appendix C. The author directed the work prompt by prompt and reviewed the AI-assisted output; machine-obtained results were verified under the protocol documented there. The author takes full responsibility for the content, including the accuracy of the references.

A The closure criterion modulo 2

Because $L \cdot L \subseteq L$ (Lemma 4.2) and the standard octonion product is $\mathbb{Z}$-bilinear, $(a + 2c) \cdot b \equiv a \cdot b \pmod{2L}$ for $a, b, c \in L$, and likewise in the right factor. The product therefore descends to an $\mathbb{F}_2$-bilinear product $\bar{\mu}$ on

$$\bar{L} := L/2L \cong \mathbb{F}_2^8, \tag{A.1}$$

the reduction of the order L modulo 2. A lattice M with $2L \subseteq M \subseteq L$ is the full preimage of $M/2L$, so it is closed under the standard octonion product exactly when $M/2L$ is a subalgebra of $(\bar{L}, \bar{\mu})$. For a twist g (Section 5.6), the lattices $g(L\bar{s})$ and $g(Ls)$ contain $2L$, since $L\bar{s} \cap Ls = 2L$ (Section 2.3) and $g(2L) = 2L$. Proposition 5.3 therefore reads modulo 2: $\star_g$ closes on Λ if and only if $g(L\bar{s})/2L$ and $g(Ls)/2L$ are subalgebras of $(\bar{L}, \bar{\mu})$.

Lagrangians and polarizations. The quadratic form

$$q\colon \bar{L} \to \mathbb{F}_2, \qquad q(v + 2L) \;:=\; N(v)/2 \pmod 2, \tag{A.2}$$

is well defined because L is even ($N(v + 2u) \equiv N(v) \pmod 4$), and its associated bilinear form, the inner product modulo 2, is nondegenerate because L is unimodular. A subspace of $\bar{L}$ is *totally isotropic* when q vanishes on it; it then lies in its orthogonal complement, so its dimension is at most 4. The 4-dimensional ones are the *Lagrangians*, and a *polarization* of $\bar{L}$ is an ordered pair of complementary Lagrangians.

Wilson's pair and σ's pair. The subspaces $L\bar{s}/2L$ and $Ls/2L$ form a polarization. They are complementary, since $L\bar{s} + Ls = L$ and $L\bar{s} \cap Ls = 2L$; each has dimension 4, since $[L : L\bar{s}] = [L : Ls] = 16$; and q vanishes on both, since $N(\ell\bar{s}) = N(\ell s) = 2N(\ell) \in 4\mathbb{Z}$ for $\ell \in L$. The isometry σ of L maps this polarization to

$$V := \sigma(L\bar{s})/2L = 2\mathcal{O}/2L, \qquad W := \sigma(Ls)/2L = p\mathcal{O}/2L, \tag{A.3}$$

where the second equalities are (4.1).[20] The identity twist does not close, since $Ls/2L$ is not a subalgebra (Remark 4.5). The twist σ closes: modulo 2, Lemma 4.3 says that V is a left ideal of $(\bar{L}, \bar{\mu})$, and Lemma 4.4 that W is a subalgebra.[21] Moreover, V is a two-sided ideal, since $2\mathcal{O}$ is one in L (Remark 4.5).[22]

Which polarizations close. The Weyl group of E_8, the isometry group of L, acts on $\bar{L}$ as the full orthogonal group of q, with kernel $\{\pm 1\}$ [WilsonBook2009, §3.12.4]. This group permutes the polarizations transitively. Indeed, each half of a polarization is its own orthogonal complement, so the associated bilinear form pairs the two halves U_1 and U_2 nondegenerately: a basis $u_1, \ldots, u_4$ of U_1 has a dual basis $u_1', \ldots, u_4'$ in U_2, and then $q\big(\sum_i (a_i u_i + b_i u_i')\big) = \sum_i a_i b_i$; the linear map between two such bases preserves q. Hence every polarization is $(g(L\bar{s})/2L, g(Ls)/2L)$ for an isometry g of L, and $\star_g$ closes on Λ exactly when both halves are subalgebras of $(\bar{L}, \bar{\mu})$.

Relation to prior work. The orthogonal space $(\bar{L}, q)$ is classical [WilsonBook2009, §3.12.4]. Every sublattice M of L isometric to $\sqrt{2}\,L$ contains $2L$, since its dual lattice, $\frac{1}{2}M$, contains L. Lepowsky and Meurman [LepowskyMeurman1982, §2] identify the lattices between $2L$ and L that are isometric to $\sqrt{2}\,L$ with the preimages of the Lagrangians, and construct a lattice isometric to Λ from any polarization; in their introduction they note that work of Tits contains what amounts to the same construction. Egan's construction (Section 6) is of this form. In the Baez series, Tim Silverman arrived at the identification of these sublattices with the Lagrangians independently (part 12 of [Baez2014]), read Egan's construction modulo 2, and found that Wilson's pair is a polarization [Silverman2016]. David Calderbank related Egan's construction to that of Lepowsky and Meurman [Calderbank2023]. The present appendix adds the product $\bar{\mu}$ inherited from the standard octonion product, under which V and W are a two-sided ideal and a subalgebra.

B Historical note: integral Cayley numbers, 1923–1946

The integral octonions this paper builds on were developed across four primary papers between 1923 and 1946. Each was obtained in the original, and its key computational content re-derived in exact arithmetic in this project (for Mahler's covering-radius theorem, by exact-arithmetic sampling).[23]

Dickson (1923) [Dickson1923] gave the first construction: an explicit 8-element basis of an order O_D for the Cayley–Dickson doubled product, with all 64 basis products closing. His Theorem XV states three maximal orders containing the eight basis units; the true count is *seven*. The cause is in his derivation (p. 321): he assumes the orders contain the Hurwitz quaternions, and exactly three of the seven do.

Kirmse (1924) [Kirmse1924] took up the problem independently. His multiplication table is itself a correct octonion algebra, and he derived a correct *necessary* form for the maximal orders, without proving sufficiency; thirty E_8-type lattices realize the form. Among the eight of those lattices that are stable under multiplication by the basis units (an unstated narrowing of the criterion, in our reconstruction), he exhibited the one, J_1 (p. 70), that is not an order. On this reading, the remaining

[20] [repo] `python_project/src/verify_polarization_lagrangians.py`.

[21] [repo] `python_project/src/verify_mod2_quotient.py`.

[22] Hence $(\bar{L}, \bar{\mu})$ is not an octonion algebra in the composition-algebra sense: those are simple.

[23] [repo] `python_project/src/verify_dickson_1923.py` [repo] `python_project/src/verify_kirmse_1924.py` [repo] `python_project/src/verify_kirmse_1924_forensic.py` [repo] `python_project/src/verify_mahler_1942.py` [repo] `python_project/src/verify_coxeter_1946.py`. Verification notes accompany each, in `paper/reviews/`.

seven of his eight are the genuine maximal orders. Kirmse also develops an ideal-theoretic treatment of the integral octonions, a substantive companion contribution that is taken up again in Mahler (1942), and methodically takes algebra alternativity as a building block, anticipating the structural role alternativity plays in later integral-octonion arithmetic.

Mahler (1942) [Mahler1942] developed the arithmetic of Dickson's order O_D: for every Cayley number X there is an integral G with $N(X - G) \leq \frac{15}{16}$ (sharper than Dickson's $\frac{5}{4}$; Mahler states without proof that $\frac{1}{2}$ is the best constant), and every left or right ideal is principal, generated by an integer multiple of an integral Cayley number of norm 1, 2, or 4.

Coxeter (1946) [Coxeter1946] identifies Kirmse's error with an explicit nonclosure witness and records that the remedy is due to Bruck: "Bruck's domain J can be derived from J_1 by transposing two of the i's" (§4). This is a transposition of two imaginary basis units, a linear involution of $\mathbb{R}^8$; the $\binom{7}{2} = 21$ such transpositions yield the seven maximal orders in seven sets of three. His §§6–13 develop the geometry of the corrected order as the E_8 lattice, the vertex set of the honeycomb 5_{21} in his notation. A postscript (§14) records his after-the-fact awareness of Dickson (1923) and Mahler (1942) and identifies Dickson's undercount with exactly the cause above.

It is curious to find that the Coxeter–Bruck transposition would prove useful here: the same linear involution of $\mathbb{R}^8$, acting now on a different object. The Coxeter–Bruck transposition acts on a maximal-order candidate that sits asymmetrically in $\mathbb{R}^8$; the present σ fixes the coordinate-symmetric placement L and does its work on the sublattice Ls of Wilson's Leech construction.

C Research methodology

This research was conducted as a collaboration between a human researcher and an AI agent (Claude, Anthropic) over a series of directed prompts, governed by a written methodology (the "Manifesto") established at the outset.

The human researcher (the author) brought the initial hypothesis (motivated by prior work on Okubo algebras and autotopies of normed composition algebras [Koeplinger2023]), established the research methodology, and maintained control of the inquiry through prompt-by-prompt review. The AI agent performed the computational work (implementing octonion algebras, Wilson's Leech lattice construction, trial algebras, and closure tests) and contributed mathematical analysis. It also drafted the manuscript text and made every revision to it, at the author's prompt-by-prompt direction.

Structural proposals from a second AI system (the Hermes agent, Nous Research, running DeepSeek V4 Flash through OpenCode) prompted the Lagrangian framing of Appendix A; the AI agent (at that stage Claude Opus 4.7) verified and corrected them.[24]

Tests were written from first principles to independently verify every foundation. This independent verification is how we know that the properties stated in the literature are correctly implemented. Closure of the twisted product was first observed on over 12,000,000 random pairs of minimal vectors with zero failures, which prompted the search for a symbolic proof.

Symbolic proof and numeric validation. The proof of Theorem 1.1 given in Section 4 is symbolic. Its finite content, the closure of $\mathcal{O}$ on a $\mathbb{Z}$-basis and the identities (4.1) on eight products each, is

[24] The proposals verbatim, the verification, and the corrections: [repo] `paper/hermes_mod2_structure_supplement.pdf`; [repo] `python_project/src/verify_discovery1_W_subalgebra.py` [repo] `python_project/src/verify_discovery2_V_isotropic.py`.

verified in exact rational arithmetic.[25] No floating-point or sampled arithmetic enters the proof. Independently, the four lemmas themselves were verified on $\mathbb{Z}$-bases: Lemma 4.1 on the eight basis vectors of L, and Lemmas 4.2 to 4.4 basis pair by basis pair. This was done in two separate computer-algebra implementations, a Python implementation using `fractions.Fraction` exact rationals and a GAP/LOOPS implementation [GAP, LOOPS] using GAP's native exact rational arithmetic.[26] This redundancy is validation, not proof: it serves to detect implementation errors, but the logical content of the proof rests on Section 4.

The methodology required:

- Every claim backed by a reference or computation.
- Every prompt logged, with actions recorded.
- Every conclusion independently verified.
- Forward-evolving corrections: no rewriting history; errors are corrected in subsequent entries.

The path to the result was not direct. Six trial algebra families were explored and ruled out before the transposition twist was identified in trial 007. The systematic methodology was essential: understanding the anatomy of failure (which conditions fail, for which vector types, and why) before trying the next approach. The failure analysis of the untwisted product (Wilson's condition (3) per Definition 2.1, exclusively Type 3×Type 3) directly pointed to the mechanism that the twist needed to correct.

Iterative convergence toward a robust result, including learning along the way what it is we are actually doing, is how study, research, and discovery have always proceeded. Mistakes were made on both sides: the AI agent occasionally departed from the agreed protocol or produced incorrect intermediate analyses, and the human author at times framed prompts under misconceptions or accepted intermediate results without sufficient scrutiny. In each case, the error was caught by the systematic cross-checking that the methodology requires (explicit verification, prompt-by-prompt review, and independent reimplementation in Python and GAP) and corrected forward. What is novel here is the ability to trace the entire path, beginning to end, through the complete list of prompts.

The full record is available at the repositories of [repo]: all prompts, code, evidence, and git history. The reader can form their own assessment of what this means for questions of discovery and attribution.

D Identities and product norms on minimal vectors

This appendix estimates, by sampling, how often the identities of Section 5.1 from left alternativity to symmetric composition hold on minimal vectors, and records the norms of products of minimal vectors. Samples were drawn uniformly from $\mathrm{Min}(\Lambda)\times\mathrm{Min}(\Lambda)$ for the binary identities, from $\mathrm{Min}(\Lambda)$ for the cube and quartic identities, and from $\mathrm{Min}(\Lambda)^3$ for the symmetric-composition identity. Sample size is $n = 10^6$ per identity; exact integer arithmetic throughout.[27] The standard error is $\mathrm{SE} = \sqrt{f(1-f)/n}$, where f is the fraction of samples on which the identity holds.

[25] [repo] `python_project/src/verify_structural_proof.py`.
[26] [repo] `python_project/src/symbolic_proof_checks.py` [repo] `python_project/src/verify_structural_proof.py` [repo] `gap_project/tests/test_lemmas.g`.
[27] [repo] `python_project/src/verify_section5_properties.py`.

Identity	Fraction of samples (% ± SE)
Left alternativity	$(0.188 \pm 0.0043)\%$
Right alternativity	$(0.188 \pm 0.0043)\%$
Flexibility	$(8.37 \pm 0.028)\%$
Cube power-associativity	$(30.70 \pm 0.046)\%$
Quartic power-associativity	$(12.75 \pm 0.033)\%$
Commutativity	$(0.018 \pm 0.0014)\%$
Norm multiplicativity	$(46.77 \pm 0.050)\%$
Symmetric composition	$(25.37 \pm 0.044)\%$

On the 10^6 samples drawn for the cube and quartic identities, every sample that satisfied the quartic identity also satisfied the cube identity.

The self-product norms $N(u \star u)$ of all 196,560 minimal vectors take exactly the values $\{16, 32, 48, 64, 80, 96, 128\}$.[28] On 10^6 sampled pairs, the pair-product norms $N(u \star v)$ took only the values $\{16, 32, 48, 64, 80, 96, 112, 128\}$, all multiples of 16, with mode 64 (the norm-multiplicative value $N(u)N(v) = 8 \times 8$).[29]

E Other twists: closure and span of the image

This appendix replaces σ by other twists (Section 5.6) and tests two properties of the resulting triple product: whether it still closes on Λ, and how much of Λ its products reach.

When $\star_g$ closes on Λ, the integer combinations of its products $u \star_g v$ $(u, v \in \Lambda)$ form a sublattice of Λ, the *span* of $\star_g$. The *span index* of $\star_g$ is the number of disjoint translates of the span that make up Λ. Span index 1 means that every vector of Λ is an integer combination of products. An order that contains a multiplicative identity always has span index 1, since every vector $u = u \cdot 1$ is a product; the ambient algebra here has no multiplicative identity, and the span index can be larger.

Both properties are decided by finitely many products. By bilinearity it suffices to take the $24 \times 24 = 576$ products of pairs of vectors from a $\mathbb{Z}$-basis of Λ. The product $\star_g$ closes on Λ if and only if all of them lie in Λ, and the span index is the index of the lattice they span, computed exactly by Hermite normal form.[30]

As an experiment, we generalize from the transposition σ to all permutations π of the seven imaginary basis units $e_1, \ldots, e_7$, each fixing e_0. These form the symmetric group S_7, with σ as a special case. For every nonidentity cycle type of S_7 we find:[31]

[28] [repo] `python_project/src/verify_idempotents_min_shell.py`.
[29] [repo] `python_project/src/verify_pair_norm_histogram.py`.
[30] [repo] `python_project/src/verify_product_span_index.py`.
[31] [repo] `python_project/src/verify_all_permutations_exact.py` [repo] `python_project/src/verify_twist_span_index_census.py`.

Cycle type of π	Permutations	Close	Rate	Span index
2-cycle	21	21	100%	2^{16}
3-cycle	70	35	50%	2^{10}
$(2,2)$	105	42	40%	2^{10}
4-cycle	210	0	0%	n/a
$(3,2)$	420	126	30%	2^{16}
5-cycle	504	252	50%	2^{10}
$(2,2,2)$	105	21	20%	2^{16}
$(3,3)$	280	112	40%	2^{10}
$(4,2)$	630	294	46.7%	2^{10}
6-cycle	840	210	25%	2^{16}
$(3,2,2)$	210	105	50%	2^{10}
$(4,3)$	420	168	40%	2^{16}
$(5,2)$	504	42	8.3%	2^{16}
7-cycle	720	336	46.7%	2^{10}
Total	5,039	1,764	35.0%	

The 2-cycles, the transpositions of Definition 3.1, close by Theorem 1.1. By Proposition 5.3, the identity permutation does not close, since Ls is not closed under the standard octonion product (Remark 4.5). Each permutation π preserves L, as σ does (Lemma 4.1), so by the same proposition the column "Close" counts the permutations π for which $\pi(L\bar{s})$ and $\pi(Ls)$ are closed under the standard octonion product. For the 3-cycles, exactly one of the two cyclic orientations of each three-element subset of imaginary indices closes.[32] No comparable pattern is given here for the other types that close in part, nor a reason for the failure of every 4-cycle.

The span index of a closing permutation depends only on its parity: it is 2^{16} for every odd one and 2^{10} for every even one (last column).

Proposition E.1. *The span of $\star$ consists of the vectors $(w,w,w)+2u$ with $w \in Ls$ and $u \in \Lambda$. In particular, $\star$ has span index $2^{16} = 65{,}536$, and its span contains 2Λ.*[33]

Proof. Write S for the span. By (4.1) and the closure of $\mathcal{O}$, $L\bar{s} \cdot_\sigma L\bar{s} \subseteq 2L\bar{s}$ and $L\bar{s} \cdot_\sigma Ls + Ls \cdot_\sigma L\bar{s} \subseteq L\bar{s}$. The products $(yp)z$ with $y,z \in \mathcal{O}$ span $\mathcal{O}$, and so do the products $y(pz)$. Indeed, $v := -\frac{1}{2}(e_1+e_2)s = \frac{1}{2}(e_0+e_2+e_3-e_7)$ lies in $\mathcal{O}$ by (2.2), as $s \in L$, and $pv+vp = 2v_0\,p - 2\langle p,v\rangle e_0 = p - e_0$, as p is purely imaginary. The right alternative law $(yx)x = yx^2$ with $x = p+v$ gives $(yp)v + (yv)p = y(pv+vp) = yp - y$, so $y = (yp)e_0 - (yp)v - \big((yv)p\big)e_0$ for every $y \in \mathcal{O}$; the left alternative law gives $y = e_0(py) - v(py) - e_0\big(p(vy)\big)$ in the same way. Now σ maps $Ls \cdot_\sigma Ls$ onto the products $(py)(pz) = p\big((yp)z\big)$ of (4.2), and $L\bar{s} \cdot_\sigma Ls$ onto the products $2y(pz)$, with $y,z \in \mathcal{O}$. Hence the products in $Ls \cdot_\sigma Ls$ span Ls, and those in $L\bar{s} \cdot_\sigma Ls$ span $L\bar{s}$.

From here on, write xy for $x \cdot_\sigma y$, while $L\bar{s}$ and Ls remain the lattices of (2.3). By the proof of Proposition 5.3, Λ is spanned by the vectors $(a,0,-a)$, $(b,-b,0)$ with $a,b \in L\bar{s}$ and $(-w,w,w)$

[32] [repo] `python_project/src/verify_twist_characterization.py`.

[33] [repo] `python_project/src/verify_product_span_index.py` [repo] `python_project/src/verify_product_span_structure.py` [repo] `python_project/src/verify_product_span_proof.py`.

with $w \in Ls$ (their sum is the vector $(a+b-w, w-b, w-a)$ there). Definition 3.3 gives

$$(a,0,-a)\star(a',0,-a') = (aa',-2aa',aa'), \qquad (b,-b,0)\star(b',-b',0) = (bb',bb',-2bb'),$$
$$(a,0,-a)\star(b,-b,0) = (2ab,-ab,-ab), \qquad (a,0,-a)\star(-w,w,w) = (-2aw,2aw,0),$$
$$(b,-b,0)\star(-w,w,w) = (-2bw,0,2bw), \qquad (-w,w,w)\star(-w',w',w') = (3ww',-ww',-ww'),$$

and the products in the opposite order are the same with the octonion factors on the right exchanged. For $c \in L\bar{s}$ and integers k_1, k_2, k_3 with $k_1+k_2+k_3 = 0$, the vector (k_1c, k_2c, k_3c) lies in Λ (Definition 2.1). So, by the inclusions above, every product other than the products of two vectors $(-w,w,w)$ lies in 2Λ. These are $2(2c,0,0) - (c,c,c)$ with $c = ww' \in Ls$, and $(2c,0,0) \in \Lambda$ as $2L \subseteq L\bar{s} \cap Ls$. Hence, by bilinearity, every vector of S has the stated form.

Conversely, S contains $(3c,-c,-c)$ for every $c \in Ls$, since the products in $Ls \cdot_\sigma Ls$ span Ls. The cyclic shift of the blocks preserves Λ and, by the $\mathbb{Z}_3$ symmetry of the routing, commutes with $\star$, so it maps S onto itself, and S also contains $(-c,3c,-c)$ and $(-c,-c,3c)$. Adding the three gives $(c,c,c) \in S$, hence $(4c,0,0) \in S$ and $2(-c,c,c) = 2(c,c,c) - (4c,0,0) \in S$. Since the products in $L\bar{s} \cdot_\sigma Ls$ span $L\bar{s}$, the products of $(a,0,-a)$ and of $(b,-b,0)$ with $(-w,w,w)$ give $2(c,-c,0)$ and $2(c,0,-c)$ in S for every $c \in L\bar{s}$. So S contains twice the vectors $(a,0,-a)$, $(b,-b,0)$ and $(-w,w,w)$ that span Λ, hence 2Λ, and with the vectors (c,c,c), $c \in Ls$, every vector of the stated form.

Finally, if $(c,c,c) \in 2\Lambda$ with $c \in Ls$, then $\frac{3}{2}c \in Ls$ by condition (3), so $\frac{1}{2}c \in Ls$. Hence $S/2\Lambda \cong Ls/2Ls$ has order 2^8, and $[\Lambda : S] = 2^{24}/2^8 = 2^{16}$. □

The transposition σ is also the reflection in the root $e_1 - e_2$ of L, and reflections in the other roots of L are twists as well. For the reflection ρ in the root $r = \frac{1}{2}(e_0 + e_1 + \cdots + e_6 - e_7)$, the triple product $\star_\rho$ closes on Λ with span index 1.[34]

The span index is invariant under ring isomorphism, since an isomorphism maps products to products and hence the span onto the span. Orders with different span indices are therefore not isomorphic. By the parity split, an order $(\Lambda, +, \star_\pi)$ with π odd (span index 2^{16}, as for σ) is not isomorphic to one with π even (span index 2^{10}), and neither is isomorphic to $(\Lambda, +, \star_\rho)$ (span index 1).

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